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```
% Model projectile motion and
% analyze how velocity and
% launch angle affect both
% range and maximum height.

% housekeeping
clc;
clear;
close all;
version
```

```
ans =

    '24.2.0.2923080 (R2024b) Update 6'
```

Project start

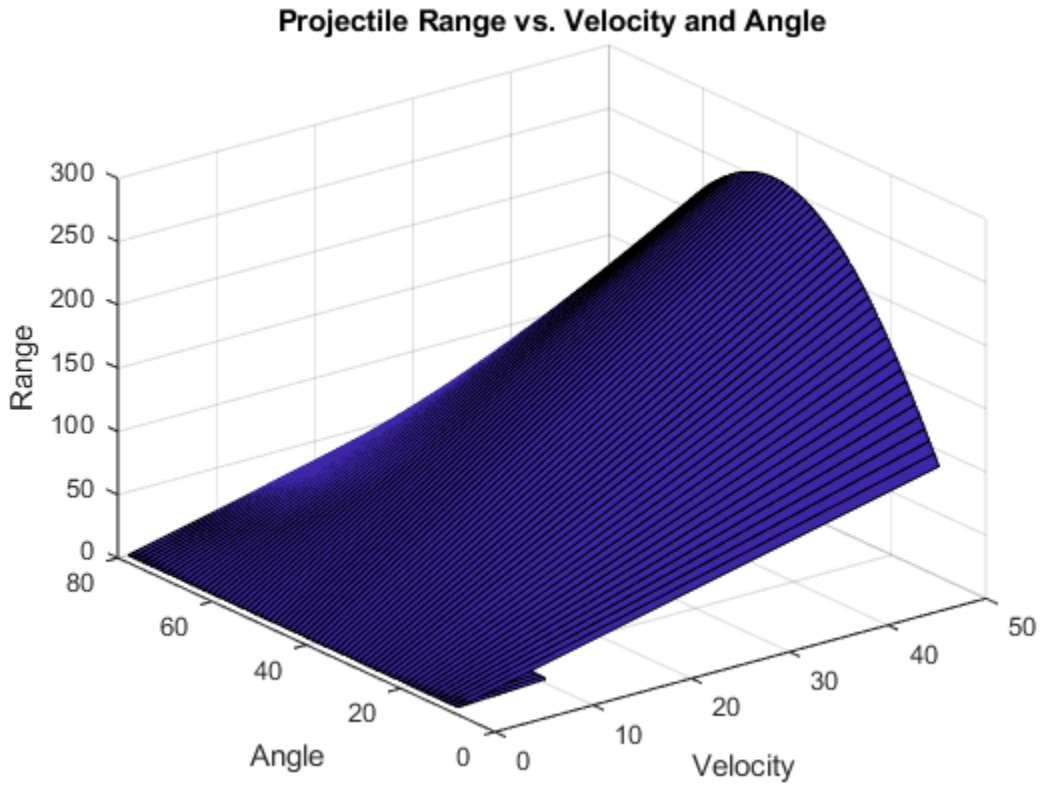
defining variables

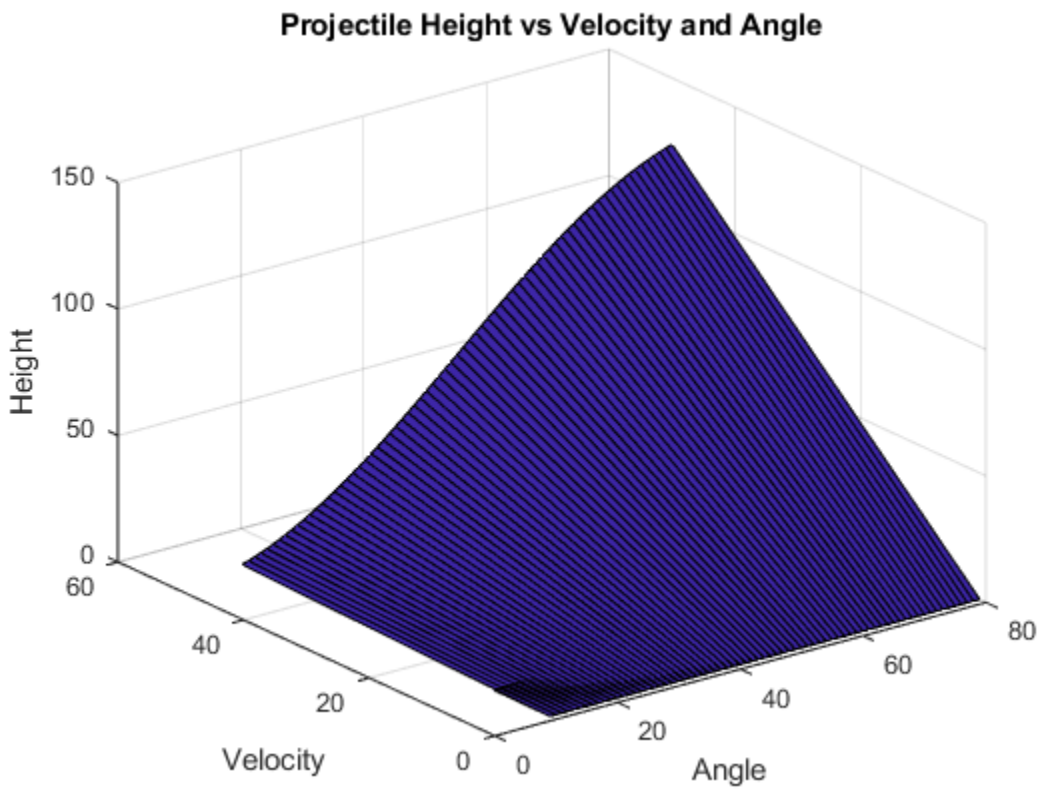
```
theta = 10:1:80;

v = [10;1;50];

g = 9.8;

% implement equation
R = (v.^2) .* sind(2*theta) / g;

% plot
figure;

surf(v, theta, R')

% labels
xlabel('Velocity')

ylabel('Angle')

zlabel('Range')

title('Projectile Range vs. Velocity and Angle')

% finding max range and location
[maxR, idxR] = max(R, [], "all");

[rowR, colR] = find(R == maxR);

maxVR = v(rowR);
```

```
maxthetaR = theta(colR);

% print the maxes

fprintf('Maximum Range = %.2f m at %.2fC and %.2f m/s\n', maxR, maxthetaR,
maxVR)

% equation determining height

H = (v.^2 .* (sind(theta)).^2) / (2*g);

% plot H results
figure;

surf(theta, v, H)

xlabel('Angle')

ylabel('Velocity')

zlabel('Height')

title('Projectile Height vs Velocity and Angle')

% find maxes and location
[maxH, idxH] = max(H, [], "all");

[rowH, colH] = find(H == maxH);

maxVH = v(rowH);

maxthetaH = theta(colH);

% print results
fprintf('Maximum Height = %.2f m at %.2fC and %.2f m/s\n', maxH, maxthetaH,
maxVH)

% while loop to find what velocity/angle causes height to exceed
vtest = 50;

maxRange = maxR;

maxHeight = maxH;

while maxRange < 1000 && maxHeight < 500
    vtest = vtest +0.01;

    rtest = (vtest^2 .* sind(2*theta)) / g;

    htest = (vtest^2 .* (sind(theta)).^2) / (2*g);

    maxRange = max(rtest);
```

```
    maxHeight = max(htest);

    if maxRange >= 1000
        limitReached = 'Range';
    elseif maxHeight >= 500
        limitReached = 'Height';
    end
end

fprintf('The maximum velocity is %.2f m/s\n', vtest)

fprintf('The maximum range is %.2f m\n', maxRange)

fprintf('The maximum height is %.2f m\n', maxHeight)

rangeLimit = 1000;

heightLimit = 500;

% if statement
if maxRange >= rangeLimit && maxHeight >= heightLimit
    fprintf('Both the range (%.0f m) and height (%.0f m) limits were
exceeded.\n', rangeLimit, heightLimit);
elseif maxRange >= rangeLimit
    fprintf('The range limit of %.0f m was reached first.\n', rangeLimit);
elseif maxHeight >= heightLimit
    fprintf('The height limit of %.0f m was reached first.\n', heightLimit);
else
    fprintf('No limits were reached.\n');
end
```

```
Maximum Range = 255.10 m at 45.00C and 50.00 m/s
Maximum Height = 123.70 m at 80.00C and 50.00 m/s
The maximum velocity is 99.00 m/s
The maximum range is 1000.10 m
The maximum height is 484.97 m
The range limit of 1000 m was reached first.
```


Projectile Range vs. Velocity and Angle
Range
300
250
200
150
100
50
0
80
60
40
20
0
Angle
0
10
20
30
40
50
Velocity


Projectile Height vs Velocity and Angle
Height
150
100
50
0
60
40
20
0
Velocity
0
20
40
60
80
Angle

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